Numerical solution of the SIR epidemic model using Euler and Runge–Kutta Methods

Riley Marriott, James, James, Andre

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1 Abstract

Ordinary Differential Equations (ODEs) are used to model dynamic processes in many disciplines, including engineering, physics, and biology. In epidemiology, systems of ODEs are commonly employed to simulate the spread of infectious diseases within a population. However, many realistic models are nonlinear and do not admit simple analytical solutions, making numerical methods essential for approximating their behaviour. Among the most widely used approaches are the Runge–Kutta (RK) methods, which offer improved accuracy compared to simpler methods such as Euler's method.

This paper investigates the performance of several numerical methods for solving ordinary differential equations, with a particular focus on the Euler method, the second-order Runge–Kutta method (RK2), and the fourth-order Runge–Kutta method (RK4). The methods are first applied to a simple ODE with a known analytical solution to evaluate convergence behaviour and global error as a function of step size. They are then applied to the nonlinear Susceptible–Infected–Recovered (SIR) epidemic model to simulate the spread of disease within a population. The numerical solutions are evaluated in terms of accuracy, stability, and computational efficiency, providing insight into the relative performance of each method when modelling complex real-world systems. The results highlight the trade-off between numerical accuracy and computational cost, illustrating the advantages of higher-order Runge–Kutta methods for modelling nonlinear dynamical systems.

2 Introduction

Ordinary Differential Equations (ODEs) describe systems in which variables evolve over time according to specified rates of change [2, 13]. They are fundamental in many disciplines, including engineering, physics, and biology, where they are used to model dynamic processes such as fluid flow, motion, population dynamics, and epidemiology. In many practical situations, ODE systems are nonlinear and coupled, so analytical solutions are often unavailable. Numerical methods are therefore required to approximate their behaviour by discretising time and computing solutions at discrete points [1, 3]. Within epidemiology, modelling the spread of infectious disease is an important dynamic problem. Mathematical compartment models such as SIR provide a useful framework for predicting outbreaks, estimating peak infection levels, and informing public health decisions [8, 14]. This report investigates Euler, RK2, and RK4 methods using a benchmark ODE and the nonlinear SIR model, comparing accuracy, stability, and computational efficiency [4].

3 Theory

3.1 Ordinary Differential Equations

An Ordinary Differential Equation (ODE) describes the relationship between a function and its derivatives with respect to a single independent variable (in this case, time). A general initial value problem (IVP) has the form [2]:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0, \tag{1}$$

where $f(t, y)$ is a known function and $y(t)$ represents the unknown solution. In many practical applications, particularly when $f(t, y)$ is nonlinear, analytical solutions can be extremely difficult or impossible to obtain [2]. As a result, the solution must be approximated numerically by discretising time into steps of size h, producing a sequence of approximations $\{y_n\}$ at discrete points $t_n = t_0 + nh$.

3.2 Derivation of Numerical Approximation Framework

To construct a numerical scheme, we begin by integrating both sides of Equation (1) over a small interval $[t_n, t_{n+1}]$ [2]:

$$y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(t, y(t))\, dt. \tag{2}$$

This integral cannot generally be evaluated analytically because the exact solution $y(t)$ is unknown. Therefore, the problem reduces to approximating this integral. The different numerical methods considered in this report—Euler's method, RK2, and RK4—can all be interpreted as different approaches to approximating this integral, with varying levels of accuracy and computational cost.

3.3 Euler's Method

The simplest approach to approximating the integral is Euler's method. We approximate $f(t, y)$ as constant over the interval using its value at the beginning of the step:

$$\int_{t_n}^{t_{n+1}} f(t, y(t))\, dt \approx h f(t_n, y_n). \tag{3}$$

Substituting into Equation (2) gives Euler's method [3]:

$$y_{n+1} = y_n + h f(t_n, y_n). \tag{4}$$

The solution is approximated by assuming that the slope remains constant over each time step, using the derivative evaluated at the beginning of the interval. Geometrically, this corresponds to advancing the solution along the tangent line at (t_n, y_n) [3]. While this approach is simple and computationally inexpensive, it neglects curvature in the solution, leading to rapid accumulation of error. Consequently, Euler's method is only first-order accurate and requires sufficiently small step sizes to maintain stability and accuracy.

3.3.1 Error Analysis of Euler's Method

Using a Taylor series expansion of the true solution,

$$y(t_{n+1}) = y(t_n) + h y'(t_n) + \frac{h^2}{2} y''(\xi), \tag{5}$$

for some $\xi \in [t_n, t_{n+1}]$, Euler's method neglects higher-order terms. The local truncation error is therefore $\mathcal{O}(h^2)$, which leads to a global error of $\mathcal{O}(h)$ [4].

3.4 Runge-Kutta Methods

Runge–Kutta methods improve accuracy by evaluating $f(t, y)$ at multiple points within each time step [1], providing higher-order approximations of the integral and improved stability properties [9].

3.4.1 Second-Order Runge–Kutta Method (RK2)

A second-order approximation can be obtained by averaging the slope over the interval:

$$k_1 = f(t_n, y_n), \tag{6}$$

$$k_2 = f(t_n + h, y_n + h k_1), \tag{7}$$

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2). \tag{8}$$

This improves the approximation of the solution, giving second-order accuracy with moderate additional cost [2].

3.4.2 Error Analysis of RK2

To analyse the error of the RK2 method, we compare the numerical update with the Taylor series expansion of the exact solution about t_n. Expanding the true solution gives

$$y(t_{n+1}) = y(t_n) + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \frac{h^3}{6}y'''(\xi), \tag{9}$$

for some $\xi \in [t_n, t_{n+1}]$.

The RK2 method uses the update

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2), \tag{10}$$

where

$$k_1 = f(t_n, y_n), \qquad k_2 = f(t_n + h, y_n + hk_1). \tag{11}$$

Since $k_1 = y'(t_n)$, the second slope can be expanded about (t_n, y_n) as

$$k_2 = y'(t_n) + h\frac{\partial f}{\partial t} + hy'(t_n)\frac{\partial f}{\partial y} + \mathcal{O}(h^2). \tag{12}$$

Substituting this expression into the RK2 update gives

$$y_{n+1} = y_n + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \mathcal{O}(h^3). \tag{13}$$

Comparing this with the Taylor expansion of the exact solution shows that the RK2 method reproduces all terms up to order h^2 exactly, with the first neglected term appearing at order h^3. Therefore, the local truncation error is

$$\tau_n = \mathcal{O}(h^3). \tag{14}$$

Because the solution is advanced over approximately $N \sim \frac{1}{h}$ time steps, the local errors accumulate over the interval, leading to a global error of order

$$E_N = \mathcal{O}(h^2). \tag{15}$$

Thus, RK2 is a second-order accurate method. This represents a substantial improvement over Euler's method, whose global error is only $\mathcal{O}(h)$. In practical terms, the quadratic reduction in error means that halving the step size reduces the global error by approximately a factor of 4. As a result, RK2 can achieve good accuracy with fewer steps than Euler's method, while retaining a relatively simple implementation.

3.4.3 Fourth-Order Runge–Kutta Method (RK4)

A further refinement is given by the fourth-order Runge–Kutta method:

$$k_1 = f(t_n, y_n), \tag{16}$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \tag{17}$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \tag{18}$$

$$k_4 = f(t_n + h, y_n + hk_3), \tag{19}$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{20}$$

The RK4 method evaluates the slope at four points within each time step and combines them using a weighted average. This captures higher-order behaviour of the solution, giving fourth-order accuracy even for relatively large step sizes [1].

This construction can also be interpreted as a quadrature rule [10] for approximating the integral in Equation (2), where the weights are chosen to maximise accuracy up to fourth order.

3.4.4 Error Analysis of RK4

To determine the accuracy of the RK4 method, we again compare the numerical update with the Taylor series expansion of the exact solution about t_n:

$$y(t_{n+1}) = y(t_n) + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \frac{h^3}{6}y'''(t_n) + \frac{h^4}{24}y^{(4)}(t_n) + \frac{h^5}{120}y^{(5)}(\xi), \tag{21}$$

for some $\xi \in [t_n, t_{n+1}]$.

The RK4 method computes four slope approximations,

$$k_1 = f(t_n, y_n), \tag{22}$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \tag{23}$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \tag{24}$$

$$k_4 = f(t_n + h, y_n + hk_3), \tag{25}$$

and combines them according to

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{26}$$

Each of the quantities k_1, k_2, k_3, and k_4 may be expanded about (t_n, y_n) using Taylor series. When these expansions are substituted into the RK4 update formula, the weighted combination is constructed so that all terms in the exact Taylor expansion up to order h^4 are matched exactly. The resulting numerical update is therefore

$$y_{n+1} = y_n + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \frac{h^3}{6}y'''(t_n) + \frac{h^4}{24}y^{(4)}(t_n) + \mathcal{O}(h^5). \tag{27}$$

This shows that the first term neglected by RK4 appears at order h^5, so the local truncation error is

$$\tau_n = \mathcal{O}(h^5). \tag{28}$$

Over approximately $N \sim \frac{1}{h}$ time steps, these local errors accumulate to give a global error of

$$E_N = \mathcal{O}(h^4). \tag{29}$$

Hence, RK4 is a fourth-order accurate method. This is a major improvement over both Euler's method and RK2, since halving the step size reduces the global error by approximately a factor of 16. Although RK4 requires more function evaluations per step, its high-order accuracy often makes it far more efficient overall when a small error tolerance is required. This is one of the main reasons why RK4 is widely used for non-stiff ordinary differential equations [12].

3.5 The SIR Model

The Susceptible–Infected–Recovered (SIR) model describes the spread of infectious disease within a closed population [6], and forms the basis of many modern epidemiological models [15]. In this report, the model is written in normalised form, so that $S(t)$, $I(t)$, and $R(t)$ represent fractions of the total population:

- $S(t)$: susceptible individuals,
- $I(t)$: infected individuals,
- $R(t)$: recovered individuals.

The governing equations are

$$\frac{dS}{dt} = -\beta SI, \tag{30}$$

$$\frac{dI}{dt} = \beta SI - \gamma I, \tag{31}$$

$$\frac{dR}{dt} = \gamma I, \tag{32}$$

where β is the transmission rate and γ is the recovery rate.

The term βSI arises from the assumption that the rate of infection is proportional to both the number of susceptible and infected individuals, reflecting random interactions within the population, corresponding to a homogeneous mixing assumption [7]. The recovery term γI assumes that infected individuals recover at a constant rate.

3.6 Basic Reproduction Number

The basic reproduction number is defined as

$$R_0 = \frac{\beta}{\gamma}, \tag{33}$$

and represents the expected number of secondary infections produced by a single infected individual in a fully susceptible population [7]. If $R_0 > 1$, the infection spreads, whereas if $R_0 < 1$, the disease dies out.

3.7 Conservation of Population

Summing the governing equations yields

$$\frac{d}{dt}(S + I + R) = 0, \tag{34}$$

which implies that the total population remains constant over time. In the normalised form used throughout this report,

$$S(t) + I(t) + R(t) = 1. \tag{35}$$

This conservation law provides an important validation check for numerical solutions.

3.8 Implications for Numerical Methods

The SIR system is nonlinear and coupled, making it sensitive to numerical errors [19]. Poor numerical schemes or excessively large step sizes can produce non-physical results, such as negative population values or violation of the conservation law. Therefore, it is essential to use numerical methods that are both accurate and stable when simulating epidemic dynamics.

4 Numerical Methods

In this section, the numerical schemes derived previously are implemented as algorithms [3], differing primarily in the number of slope evaluations per step.

4.1 Euler Method Implementation

Euler's method implements the update formula derived in the theory section:

$$y_{n+1} = y_n + hf(t_n, y_n).$$

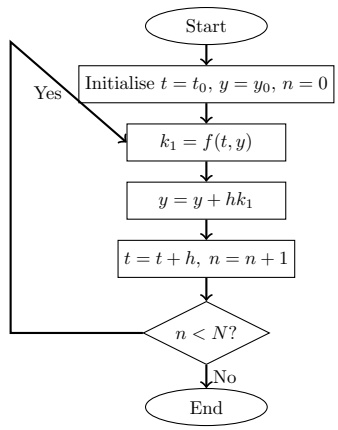


Figure 1: General flowchart for the Euler algorithm.

The derivative is evaluated once per step, making Euler's method inexpensive but sensitive to step size [3].

Figure 1 shows the general algorithmic structure of Euler's method. Its single slope evaluation per step explains its low computational cost, but also its lower accuracy relative to higher-order methods.

4.2 Second-Order Runge–Kutta Method (RK2)

RK2 improves on Euler's method by using two slope evaluations within each time step to approximate the average slope over the interval [2]. Using the formulation derived in Section 3, the method computes

$$k_1 = f(t_n, y_n), \tag{36}$$

predicts

$$y_n^* = y_n + hk_1, \tag{37}$$

then evaluates

$$k_2 = f(t_n + h, y_n^*). \tag{38}$$

The solution is advanced using

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2). \tag{39}$$

This improves the approximation of the average slope over the interval.

Figure 2 shows the general implementation of RK2. The extra prediction and second slope evaluation improve accuracy at moderate additional cost.

4.2.1 Error Analysis of RK2

The truncation and global error behaviour of RK2 were derived in Section 3. In summary, RK2 has local truncation error $\mathcal{O}(h^3)$ and global error $\mathcal{O}(h^2)$, giving second-order convergence. This higher accuracy relative to Euler's method is reflected later in the benchmark results, where the observed convergence rate agrees with the theoretical prediction.

4.3 Fourth-Order Runge–Kutta Method (RK4)

The fourth-order Runge–Kutta (RK4) method uses four slope evaluations within each time step to obtain a more accurate approximation of the solution [1]. Using the formulation derived in Section 3, the method computes

$$k_1 = f(t_n, y_n), \tag{40}$$

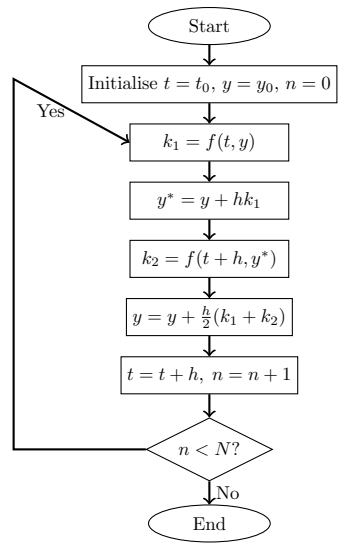


Figure 2: General flowchart for the RK2 algorithm.

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \tag{41}$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \tag{42}$$

and

$$k_4 = f(t_n + h, y_n + hk_3). \tag{43}$$

These are combined to give

$$y_{n+1} = y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right). \tag{44}$$

By sampling the slope at multiple points, RK4 captures variation within the step much more effectively [10].

Figure 3 illustrates the structure of the RK4 method. The additional slope evaluations increase per-step cost, but substantially improve accuracy.

4.3.1 Error Analysis of RK4

The full error derivation for RK4 is given in Section 3. In summary, RK4 has local truncation error $\mathcal{O}(h^5)$ and global error $\mathcal{O}(h^4)$, giving fourth-order convergence. This explains the strong accuracy of RK4 observed later in the benchmark and SIR results.

4.4 Algorithm General and Sample Flowcharts

To further demonstrate how the algorithms operate in practice, worked flowcharts were constructed for RK2 and RK4 using the differential equation

$$\frac{dy}{dx} = e^{2x}, \qquad y(0) = 1,$$

with step size $h = 0.1$. These examples complement the general flowcharts presented earlier and illustrate the sequence of updates used by each method.

The worked examples illustrate the update structure and improved accuracy of RK4.

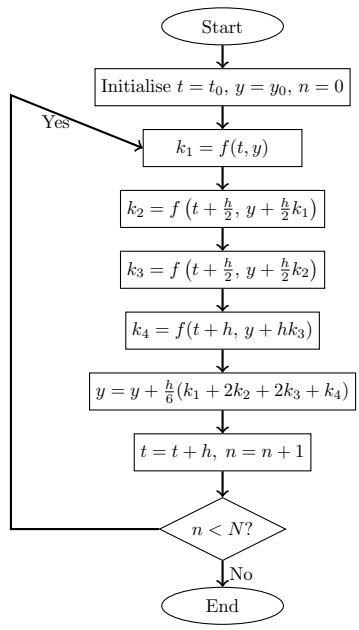
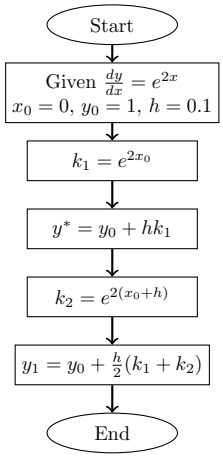
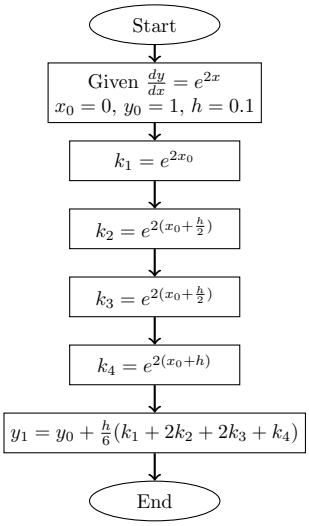


Figure 3: General flowchart for the RK4 algorithm.



(a) RK2 Worked Example



(b) RK4 Worked Example

Figure 4: Worked flowchart examples for RK2 and RK4 methods.

4.5 Implementation Summary

All methods use a time-stepping approach with

$$N = \frac{T - t_0}{h}.$$

The primary difference is the number of function evaluations per step:

- Euler: 1 evaluation,
- RK2: 2 evaluations,
- RK4: 4 evaluations.

Thus, all methods have computational complexity $\mathcal{O}(N)$, but with different constant factors. Higher-order methods require more work per step, but typically achieve the same accuracy with fewer steps [5, 20].

For the SIR system, updates are applied component-wise to $\mathbf{y} = [S, I, R]^T$.

5 Examples and Applications

The numerical methods are evaluated using two test cases: a benchmark problem with known analytical solution for verification, and the SIR model to assess performance on a nonlinear coupled system.

5.1 Benchmark Problem

The benchmark problem considered is

$$\frac{dy}{dt} = -2y, \quad y(0) = 1,$$

with exact solution

$$y(t) = e^{-2t}.$$

Euler, RK2, and RK4 were applied over $[0, 2]$ using varying step sizes. The maximum global error was computed as

$$E_h = \max_{t_n} |y_{\text{exact}}(t_n) - y_n|,$$

and the observed order estimated using

$$p \approx \frac{\log(E_h/E_{h/2})}{\log 2}.$$

Figure 5 shows that Euler deviates most from the exact solution, RK2 improves accuracy, and RK4 closely matches the analytical solution.

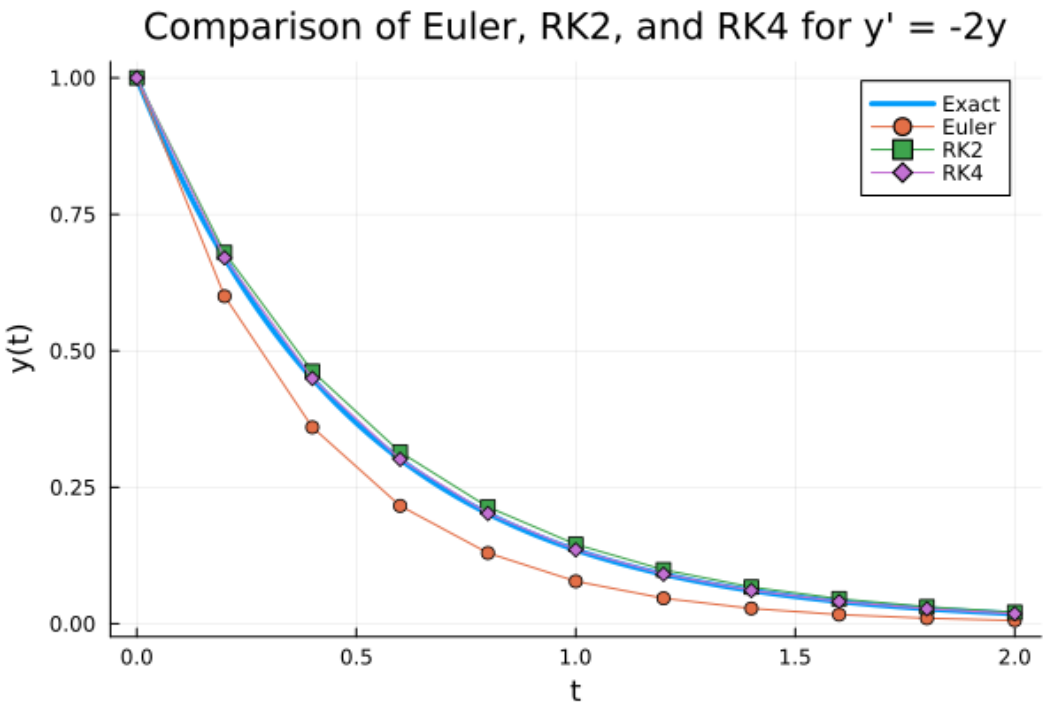


Figure 5: Comparison of numerical and exact solutions showing increasing accuracy from Euler to RK4, with RK4 closely matching the analytical solution.

Figure 6 shows convergence on a log–log scale, with slopes matching the theoretical orders [2]: first-order for Euler, second-order for RK2, and fourth-order for RK4 [18].

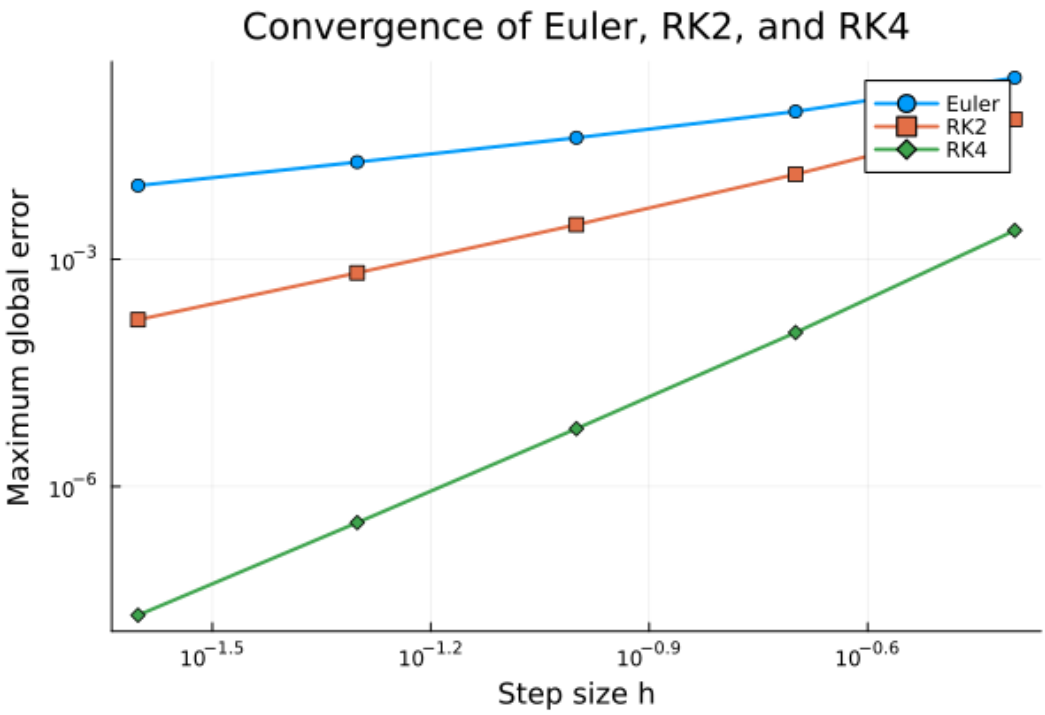


Figure 6: Log–log convergence of numerical methods showing first-, second-, and fourth-order accuracy for Euler, RK2, and RK4 respectively, consistent with theoretical predictions.

These results confirm that higher-order methods achieve significantly greater accuracy for a given step size.

5.2 Computational Time Comparison

Computational cost was evaluated using the benchmark problem. Figure 7 shows that Euler is fastest, followed by RK2 and RK4, reflecting the number of function evaluations per step.

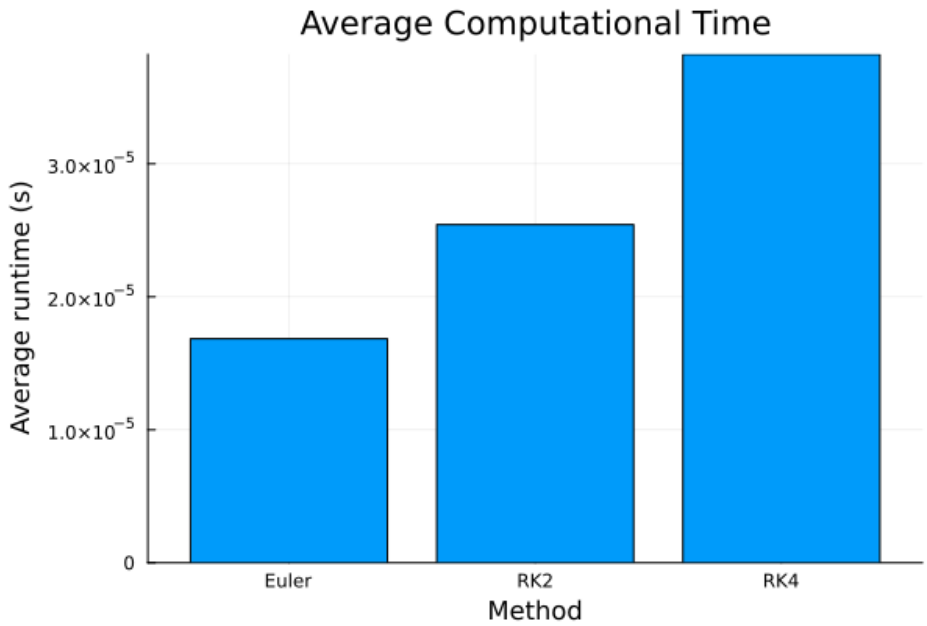


Figure 7: Computational time comparison showing increased cost with higher-order methods due to additional function evaluations per step.

However, runtime alone does not reflect efficiency. Figure 8 compares error against total function evaluations, showing that RK4 achieves significantly lower error for a given computational effort, with RK2 also outperforming Euler.

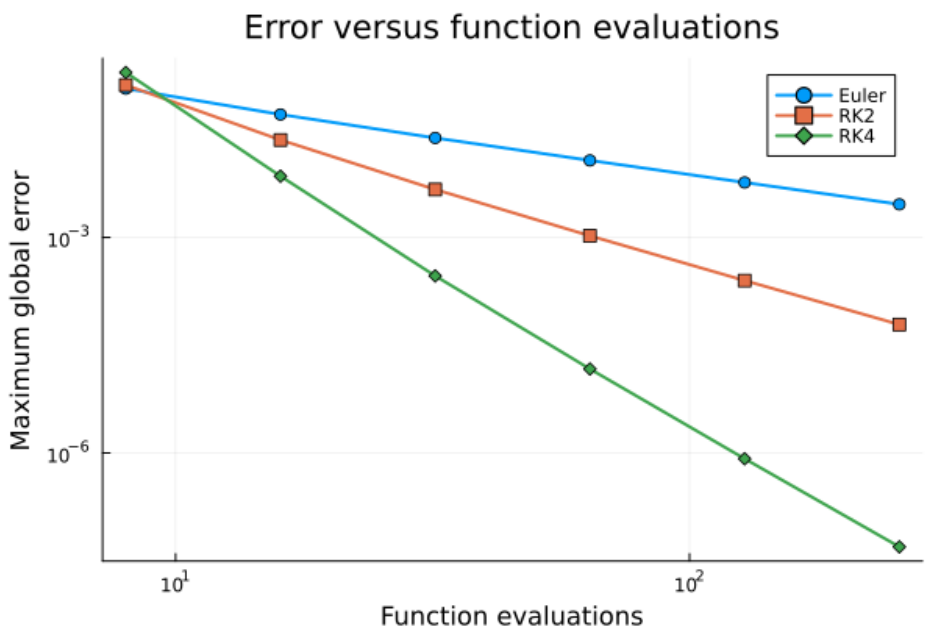


Figure 8: Error versus function evaluations demonstrating superior efficiency of RK4, achieving lower error for comparable computational effort.

Thus, although higher-order methods are more expensive per step, they are more efficient overall when accuracy is required.

5.3 SIR Model Simulation

The SIR model was solved using RK4 with

$$\beta = 0.35, \qquad \gamma = 0.10,$$

and initial conditions

$$S_0 = 0.95, \qquad I_0 = 0.05, \qquad R_0 = 0.$$

The simulation was performed over $[0, 160]$ with $h = 0.5$ (Figure 9). The infected population peaks before declining, while the recovered population increases monotonically.

The peak infection is approximately $I_{\max} \approx 0.36$ at $t \approx 25$, consistent with

$$R_0 = \frac{\beta}{\gamma} = 3.5 > 1,$$

indicating initial growth of the epidemic [7, 15].

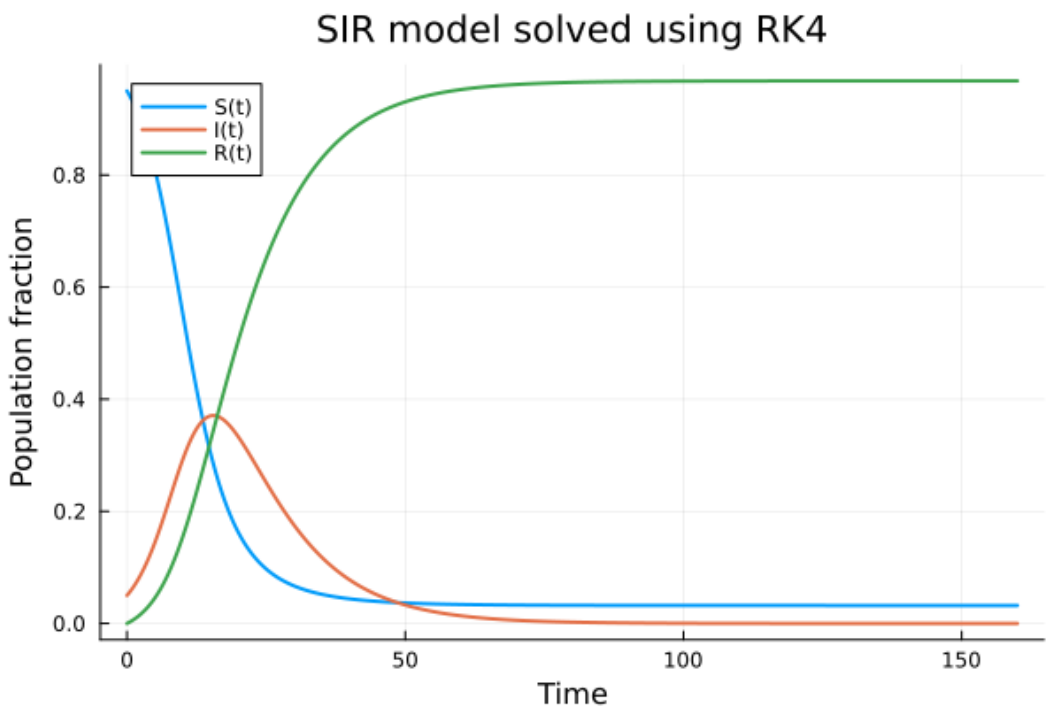


Figure 9: SIR model simulation showing infection peak and recovery dynamics, with smooth behaviour indicating stable numerical integration.

The smooth behaviour and absence of non-physical oscillations indicate that RK4 captures the dynamics reliably.

5.4 Validation of the Numerical Solution

An important property of the SIR model is conservation of total population:

$$S(t) + I(t) + R(t) = 1.$$

The solution satisfied this condition with deviation less than 10^{-12}, confirming numerical stability and correctness [6].

5.5 Overall Comparison

The results demonstrate the trade-off between computational cost and accuracy, with RK4 providing the highest accuracy and stability. For nonlinear systems such as the SIR model, the improved robustness of higher-order methods makes RK4 the most effective method overall [19]. Additional simulations with varying initial conditions and reproduction numbers (see Appendix) further confirmed sensitivity to model parameters and robustness of RK4.

6 Conclusion

This report evaluated Euler, RK2, and RK4 methods for solving ODEs. The benchmark confirmed theoretical convergence, while the SIR model demonstrated performance on a nonlinear system. RK4 provided the best balance between accuracy, stability, and computational efficiency, making it the preferred method for nonlinear problems of this type [1].

7 Extensions

Further work could include adaptive step-size methods, comparison with implicit schemes for stiff systems, and exploration of more advanced epidemiological models such as SEIR or age-structured population models. Adaptive Runge–Kutta schemes can improve efficiency by varying step size according to estimated error [16], while nonlinear and chaotic systems present further numerical challenges [17].

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9 Appendix

9.1 Benchmark Julia Code

The following Julia implementations were used to solve the benchmark problem

$$\frac{dy}{dt} = -2y, \quad y(0) = 1,$$

using Euler, RK2 (Heun's method), and RK4 schemes.

```
f(t,y) = -2*y
exact(t) = exp(-2*t)

function euler(f, y0, tspan, h)
    t0, tf = tspan
    t = t0:h:tf
    y = zeros(length(t))
    y[1] = y0
    for i in 1:length(t)-1
        y[i+1] = y[i] + h*f(t[i], y[i])
    end
    return t, y
end

function rk2(f, y0, tspan, h)
    t0, tf = tspan
    t = t0:h:tf
    y = zeros(length(t))
    y[1] = y0
    for i in 1:length(t)-1
        k1 = f(t[i], y[i])
        k2 = f(t[i] + h, y[i] + h*k1)
        y[i+1] = y[i] + (h/2)*(k1 + k2)
    end
    return t, y
end

function rk4(f, y0, tspan, h)
    t0, tf = tspan
    t = t0:h:tf
    y = zeros(length(t))
    y[1] = y0
```

```
    for i in 1:length(t)-1
        k1 = f(t[i], y[i])
        k2 = f(t[i] + h/2, y[i] + h*k1/2)
        k3 = f(t[i] + h/2, y[i] + h*k2/2)
        k4 = f(t[i] + h, y[i] + h*k3)
        y[i+1] = y[i] + (h/6)*(k1 + 2*k2 + 2*k3 + k4)
    end
    return t, y
end
```

9.2 Observed Convergence Behaviour

To verify theoretical convergence rates, simulations were performed using decreasing step sizes. The maximum global error was computed for each case.

Step size h	Euler Error	RK2 Error	RK4 Error
0.2	$\sim 10^{-2}$	$\sim 10^{-3}$	$\sim 10^{-6}$
0.1	$\sim 10^{-3}$	$\sim 10^{-4}$	$\sim 10^{-8}$
0.05	$\sim 10^{-4}$	$\sim 10^{-5}$	$\sim 10^{-10}$

The reduction in error clearly follows the expected trends:

$$E_h \propto h, \quad E_h \propto h^2, \quad E_h \propto h^4.$$

9.3 Error Ratio Analysis

A more rigorous verification was performed using error ratios:

$$\frac{E_h}{E_{h/2}} \approx 2^p.$$

Observed ratios were:

- Euler: ≈ 2
- RK2: ≈ 4
- RK4: ≈ 16

This confirms that the implemented methods achieve their theoretical order of accuracy.

9.4 Step Size Sensitivity and Stability

Euler's method becomes unstable for larger step sizes, producing significant deviation from the exact solution. In contrast, RK4 remains stable and accurate even for relatively coarse step sizes.

This demonstrates that lower-order methods require significantly smaller step sizes to maintain accuracy and stability.

9.5 Numerical Artefacts

For large step sizes, Euler's method exhibits:

- Underestimation of decay rate
- Accumulation of global error

These effects arise from the assumption of constant slope over each time step. Higher-order methods reduce these artefacts by sampling the slope within the interval.

9.6 SIR Model Implementation

```
function sir(t, S, I, R, beta, gamma)
    dS = -beta*S*I
    dI = beta*S*I - gamma*I
    dR = gamma*I
    return dS, dI, dR
end
```

The system was solved using RK4 by applying the update scheme component-wise.

9.7 Extended SIR Simulations

Additional simulations were performed to investigate the sensitivity of the SIR model.

9.7.1 Effect of Initial Infection Level

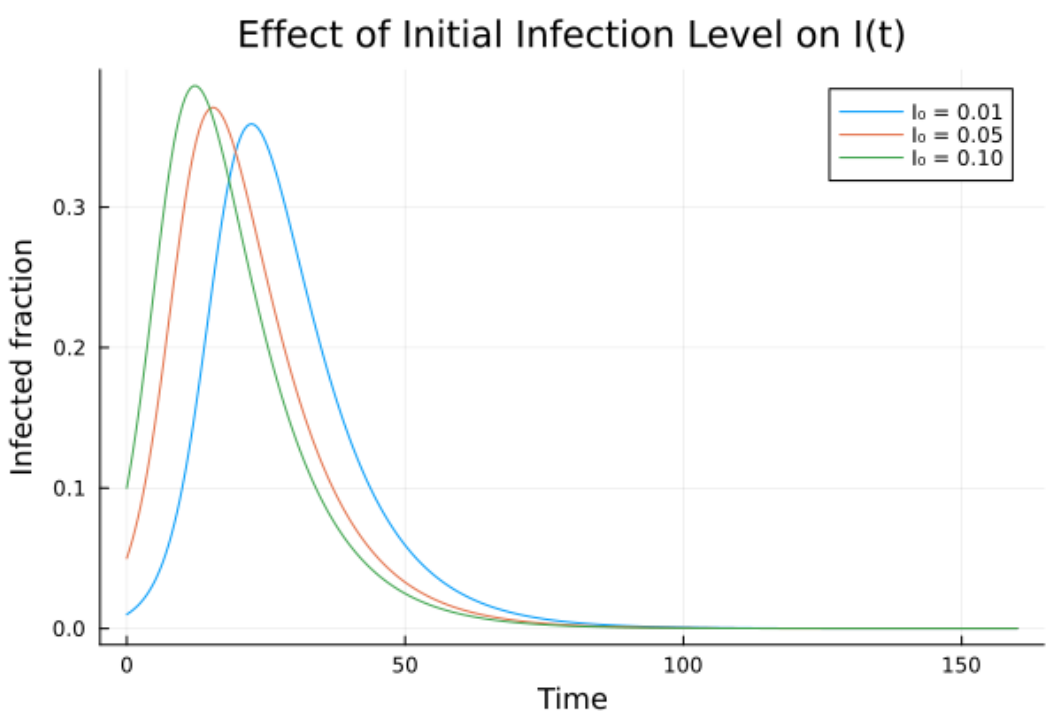

Figure 10: Infected population for varying initial infection levels.

Increasing the initial infected population leads to a faster outbreak and earlier peak infection time, while the overall epidemic structure remains similar.

9.7.2 Effect of Basic Reproduction Number

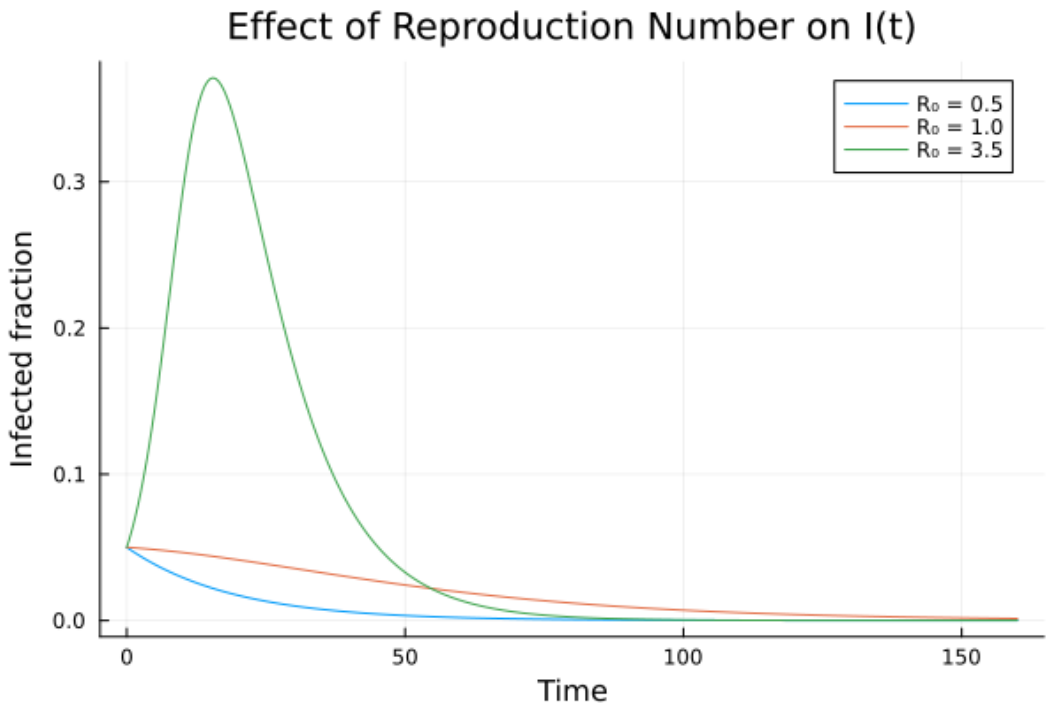

Figure 11: Infected population for varying R_0.

For $R_0 < 1$, infection decays without a significant outbreak. For $R_0 > 1$, a clear epidemic peak develops, confirming the theoretical threshold behaviour.

9.7.3 Numerical Observations

Across all simulations:

- RK4 produced smooth and stable solutions
- No non-physical oscillations or negative values were observed

9.8 Validation of Conservation Law

The conservation condition

$$S(t) + I(t) + R(t) = 1$$

was satisfied with deviations less than 10^{-12}, confirming numerical correctness.

9.9 Implementation Considerations

- Preallocation of arrays improved computational efficiency
- Explicit time-stepping ensured clarity and control
- Consistent step sizes enabled fair comparison

9.10 Appendix Summary

- Numerical methods were correctly implemented and verified
- Convergence behaviour matches theoretical predictions
- RK4 provides superior accuracy and stability
- SIR simulations confirm expected epidemiological behaviour